Nucleophilic Substitution and Elimination

Ingyu Bahng

DATE PLACEHOLDER

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1 Nucleophilic Substitution

After generating a haloalkane compound via radical halogenation (*as covered in Radical Halogenation*), the next step is to substitute that halogen with a different atom/molecule of our choosing. One of the methods we can use to carry this out is nucleophilic substitution.

Core Concept 1.1: Nucleophilic Substitution

Nucleophilic substitution is a fundamental class of organic reactions where an e^- rich species, called a **nucleophile (Nu)**, attacks an electron-deficient atom called the **electrophile (E)**, displacing a specific group known as the **leaving group (L)**, which departs with the bonding electron pair. Halogens are common leaving groups due to their electronegativity and ability to stabilise the negative charge upon departure.

Nucleophilic substitution can be categorized into S_N2 and S_N1 .

Core Concept 1.2: S_N2

S_N2 reactions are **bimolecular**, meaning that both the leaving group and nucleophile are present in the transition state and making the reaction rate dependent on both [Nu] and [E-L] (Rate $= k[CH_3Cl][^-OH]$). The following is a chemical reaction of CH_3Cl undergoing S_N2 reaction with OH^-.

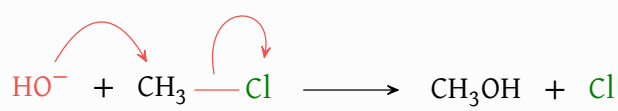

Figure 1: S_N2 equation

Here, Cl is the leaving group and OH^- is the nucleophile. From a 3D point of view, we see the following outline interaction.

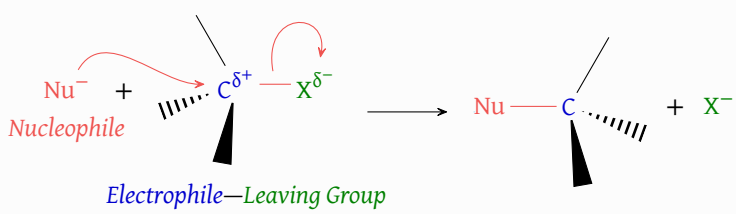

Figure 2: S_N2 mechanism reaction

It is important to note that S_N2 reactions occur in **one step**, where bond making and breaking occur simultaneously, meaning there are no intermediates — unlike the two-step propagation of radical halogenation (EXTERNAL REFERENCE TO RAD HAL). Consequently, the energy diagram consists of a single hill and is exothermic when the nucleophile is a more favorable attachment than the leaving group. *L is denoted as X in the figure below.*

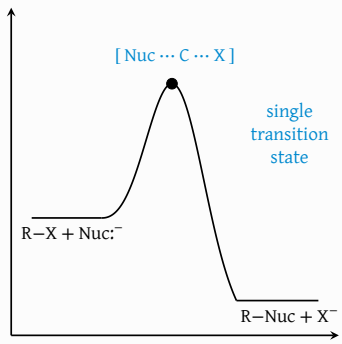

Figure 3: S_N2 energy diagram

A result of the reaction being one step is **stereospecificity** — there is *always* an inversion. A few key consequences of this are worth keeping in mind:

(a) Inversion does not always imply a shift between the R and S configuration (EXTERNAL REFERENCE TO CONFORMATIONAL ANALYSIS)

(b) Two successive S_N2 reactions result in structural retention (double inversion)

(c) Performing S_N2 on only one chiral center of a molecule with two chiral centers yields a diastereoisomer (a non-mirror-image stereoisomer)

(d) Starting with a **racemate** (a 50/50 mix of R and S enantiomers) will still yield a racemate, as all configurations are inverted

All of these points follow intuitively from the structural consequences of the S_N2 mechanism, though they may not be immediately obvious.

Before exploring S_N1 reactions, it is worth establishing what defines a **"good" leaving group and nucleophile**, as this governs the outcome of both mechanisms.

Core Concept 1.3: Leaving Group Quality

There are two factors that determine whether a leaving group is good: **electronegativity** and **orbital size**. Regarding electronegativity, a good leaving group is also a weak base A^- — one that is stable with an extra e^- due to its high electronegativity. Regarding orbital size, a larger orbital size element forms relatively weaker bonds, as its molecular bonding orbitals occur at higher energy levels compared to smaller elements, making it more inclined to dissociate. Together, these two factors mean that better leaving groups appear moving left to right (→) and top to bottom (↓) across the periodic table.

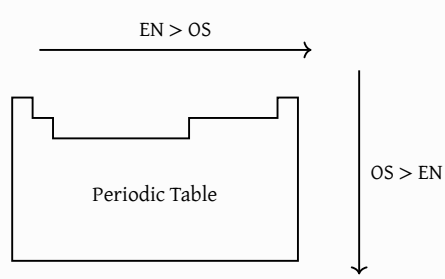

Figure 4: Leaving group ability within the periodic table

Comparing down the periodic table for halogens:

$$F^- < Cl^- < Br^- < I^-$$

I^- is the best leaving group due to its large orbital size and correspondingly weak bond. Notably, I^- is such a great leaving group that it does not even participate in radical halogenation, making it impractical in that context. F^-, on the other hand, has the smallest orbital size, forming much stronger bonds through lower energy orbitals, making it the poorest leaving group of the halogens.

Comparing across the periodic table (*row 2*):

$$-CH_3 < -NH_2 < -OH < -F$$

Here, F^- is the best leaving group due to its greater electronegativity relative to O, N, and C, allowing it to better stabilise the extra e^- upon departure. In practice, however, F^- remains a poor leaving group relative to the broader periodic table, HO^- is exceedingly poor, and $NH_2{}^-$ and $CH_3{}^-$ do not leave at all.

For **polyatomic** leaving groups, **resonance** (π conjugation) plays an important additional role. The ability to delocalise e^-s across an adjacent p orbital system allows the leaving group to absorb excess charge more effectively — a larger delocalization system not only stabilises the extra charge but also weakens the leaving group's basicity, making it a better leaving group overall.

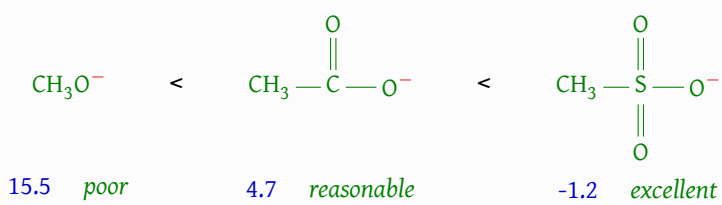

Figure 5: Polyatomic leaving groups; *corresponding numbers are pK_a values*

Core Concept 1.4: Nucleophile Quality

Nucleophilicity depends on four factors: **charge-basicity**, **basicity**, **solvation**, and **polarizability**.

(a) **Charge-basicity** refers to the greater tendency of an anion, relative to its neutral counterpart, to form a bond with a cation in order to delocalise its net negative charge.

$$OH^- > H_2O \qquad NH_2{}^- > NH_3 \qquad SO_4{}^{2-} > ROSO_3{}^-$$

(b) **Basicity** refers to how readily the nucleophile donates e^- density to the electrophile — analogous to how it would donate to H^+. As electronegativity decreases moving right to left across the periodic table, nucleophilicity increases accordingly.

$$H_3N > H_2O \qquad H_2N^- > HO^- \qquad HO^- > F^-$$

(c) **Solvation** refers to the ability of the surrounding solvent to impede the nucleophile's ability to attack the electrophile. In **polar protic solvents** (*such as water or ethanol, which contain O-H and N-H bonds*), strong hydrogen bonds effectively "handcuff" the nucleophile. Larger nucleophiles are less susceptible to this because their diffuse e^- clouds result in lower charge densities, weakening the solvent's grip.

$$F^- < Cl^- < Br^- < I^-$$

In **polar aprotic solvents** (*polar solvents lacking O-H and N-H bonds*), hydrogen bonding is absent and the

nucleophile is largely unimpeded. The trend therefore reverses to reflect intrinsic basicity.

$$I^- < Br^- < Cl^- < F^-$$

Note that F^- remains a weak base in the grand scheme — it is simply less solvated than I^- in protic solvents, and does not compare to stronger bases such as OH^- or $NH_2{}^-$.

(d) **Polarizability** refers to the large, diffuse e^- clouds of heavier elements, which allow for more effective orbital overlap in the S_N2 transition state.

$$H_2O < H_2S < H_2Se \qquad \text{or} \qquad (CH_3)_3N < (CH_3)_3P$$

Nucleophilicity arises from a complex interplay of these factors. At its core, it is driven by **basicity**, which governs the horizontal periodic trend — atoms become less electronegative and more willing to donate e^- density moving left. The vertical trend, however, is governed by the solvent. In **polar protic solvents**, smaller nucleophiles such as F^- are heavily solvated, making larger, less-impeded ions like I^- the stronger attackers. In **polar aprotic solvents**, these hydrogen-bonding impediments are absent, allowing the intrinsically more basic smaller ions to reassert themselves as the stronger nucleophiles. **Polarizability** provides larger elements an additional advantage in S_N2 transition states, as their more diffuse e^- clouds can begin overlapping with the electrophilic carbon's orbital from a greater distance.

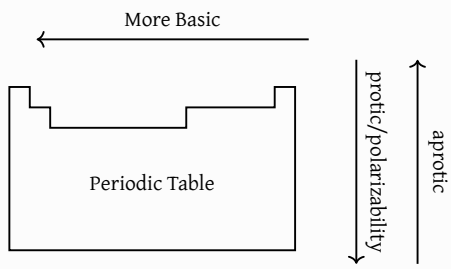

Figure 6: Nucleophilic ability within the periodic table

Core Concept 1.5: Steric Effects on Nucleophilic and Leaving Group Ability

Steric effects must be considered for both the nucleophile and leaving group. Steric effects arise from e^- repulsion between the nucleophile or leaving group and the three substituents bonded to the target electrophile.

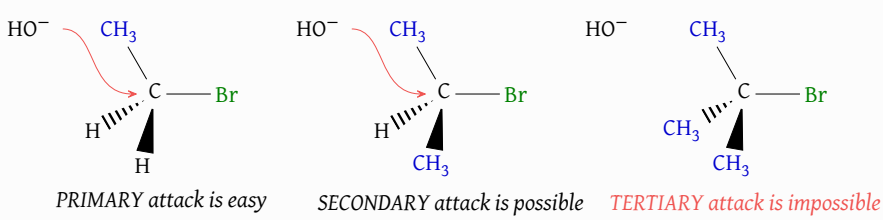

Figure 7: Steric effects in the context of S_N2 reactions

As a consequence, **leaving groups are better when larger**, as the repulsion assists their departure, while **nucleophiles are better when smaller**, as bulk impedes their approach. This steric hindrance is greatest when substituents are directly on the electrophilic carbon (α) and diminishes with distance (β, γ, δ, etc.).

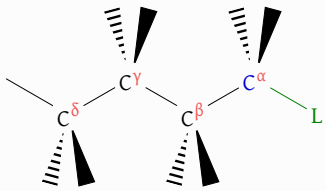

Figure 8: Carbon branching

Branch	**Structure & Relative Rate**			
	Hydrogens	Primary	Secondary	Tertiary
α	145	1	0.028	0
β	1	0.8	0.05	10^{-5}

Table 1: Rate of S_N2 reactions across different branching patterns

Core Concept 1.6: S_N1

S_N1 reactions are **unimolecular**, meaning the reaction rate is solely determined by [R – L].

$$\text{reaction rate} = k[\text{R}-\text{L}]$$

Unlike S_N2, S_N1 proceeds through a series of distinct steps.

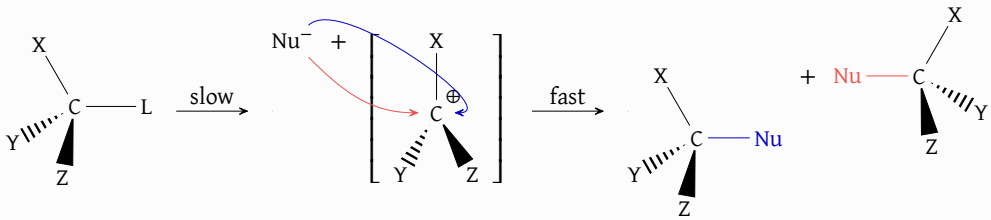

Figure 9: S_N1 mechanism

The first step — spontaneous dissociation of the leaving group from the electrophile — is the **rate limiting step**, as it is highly unfavourable, forming only trace amounts of carbocations at equilibrium. This makes it the bottleneck of the reaction, after which the nucleophile attacks either face of the sp^2-hybridised electrophile's perpendicular p orbitals.

Several important consequences follow from this mechanism.

(a) S_N1 nucleophilic attack is **not stereospecific**. A stereochemically pure reactant pool will yield a mixture of stereoisomers upon S_N1. In practice, **slight inversion** is observed due to the transient physical blockade created by the lingering leaving group, which remains in proximity to the carbocation p orbital through ionic attraction.

(b) Because the rate is limited solely by carbocation formation, the **rate increases with a better leaving group**.

$$-\text{Cl} < -\text{Br} < -\text{I} < -\text{O}_3\text{SR}$$

(c) S_N1 **is accelerated by increasingly polar solvents**, particularly polar protic solvents. As the leaving group

begins to depart, polar solvent molecules stabilise it through hydrogen bonding, effectively lowering the energy barrier for its departure.

$$(CH_3)_3CBr \xrightarrow{100\%\ H_2O} (CH_3)_3COH + HBr \qquad 400,000$$

$$(CH_3)_3CBr \xrightarrow[10\%\ H_2O]{90\%\ \text{acetone}} (CH_3)_3COH + HBr \qquad 1$$

Figure 10: Relative S_N1 reaction rates in polar solvents

(d) **Product-determining steps occur after the bottleneck.** Once the carbocation intermediate forms, the dominant product is determined by nucleophilic competition — the strongest nucleophile in solution will attack preferentially. Consider, for instance, a mixture of NaCl and NaI in acetone.

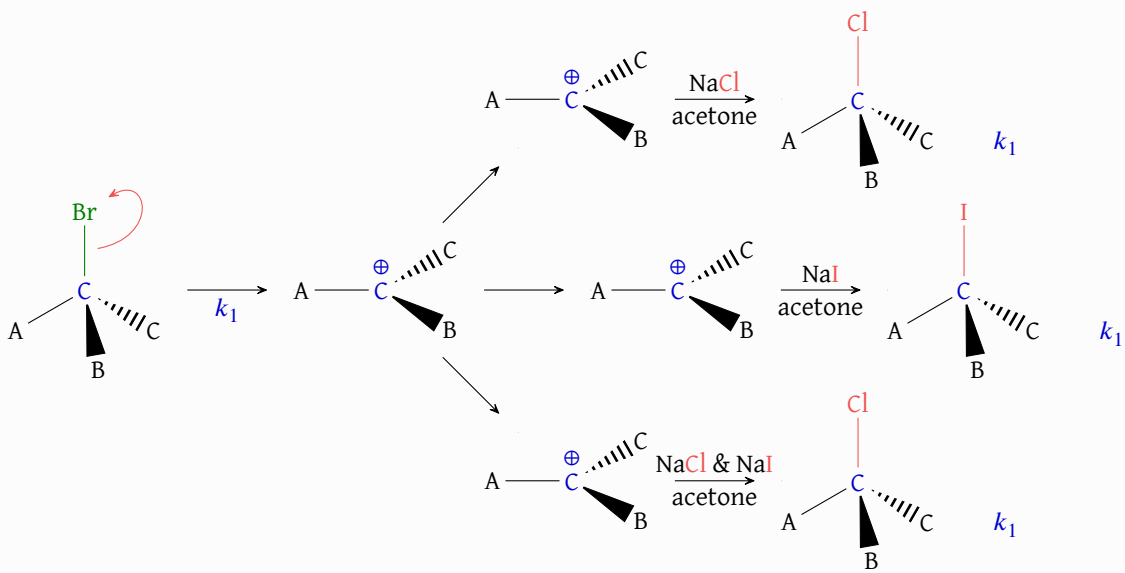

Figure 11: S_N1 nucleophile competition; *Cl is a better nucleophile than I, thus a mixture will result in the Cl product (bottom reaction)*

Here, the overall reaction rate is identical across all paths, as it depends solely on the departure of the leaving group (*regardless of the nucleophile identity*). Because acetone is a polar aprotic solvent, the Cl^- product will be favored under a mixture of Cl^- and I^- because Cl^- is a stronger nucleophile than I^- is.

Core Concept 1.7: Competition between S_N2 and S_N1

This raises an important question — if S_N1 and S_N2 ultimately achieve the same outcome, what drives a system to favour one over the other? Two factors determine this.

(a) α-branching disfavours S_N2 by sterically impeding nucleophilic approach, making it virtually impossible for a nucleophile to attack a tertiary electrophile in an S_N2 fashion.

(b) α-branched carbocations are stabilised by hyperconjugation with neighbouring molecular orbitals, lowering the energy of the rate-limiting step and making S_N1 viable.

$$\underbrace{H_3C^+ \text{ (methyl)} < H_3CH_2C^+ \text{ (primary)}}_{\text{energetically inaccessible in solution}} < \underbrace{(H_3C)_2HC^+ \text{ (secondary)} < (H_3C)_3C^+ \text{ (tertiary)}}_{\text{only cations feasible in solution}}$$

The table below summarises the conditions favouring S_N1 and S_N2 across different substrate classes. *Note that secondary R substrates do not inherently favour either mechanism — the outcome depends heavily on solvent and the nature of the nucleophile and leaving group.*

R	S_N1	S_N2
CH_3	**Not observed** (methyl cation too high energy)	**Frequent** (fast with good Nu/L)
Primary	**Not observed** (primary carbocation too high energy)	**Frequent** (fast with good Nu/L, slow when β branching present)
Secondary	**Relatively slow** (best with good L in polar protic solvents)	**Relatively slow** (best with high concentrations of good Nu in polar aprotic solvents)
Tertiary	**Frequent** (particularly fast in polar, protic solvents and good L)	**Extremely slow**

2 Elimination

Core Concept 2.1: Elimination

While substitution reactions involve replacing one group with another, **elimination** reactions involve **removal** — specifically, the loss of a leaving group and an adjacent hydrogen to form a new π bond, yielding an alkene. Like nucleophilic substitution, elimination reactions are classified into two types: E1 and E2.

Core Concept 2.2: Elimination 1 (E1)

Following carbocation formation in S_N1, the nucleophile can alternatively act as a base (B^-), deprotonating a neighbouring alkyl group to yield a π bonded alkene over two distinct steps. *Note that this pathway requires a neighbouring alkyl group and is therefore not possible for methane.*

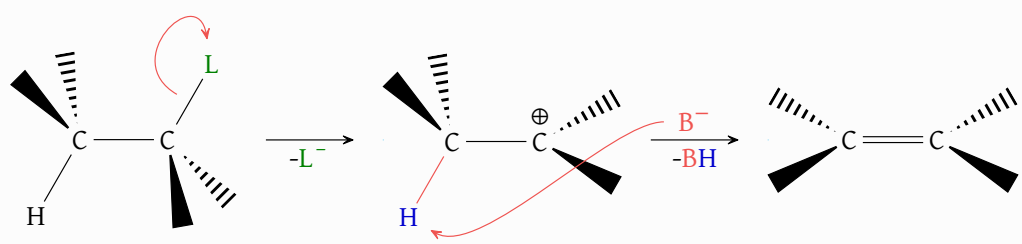

Figure 12: E1 mechanism

The rate-determining step is identical to that of S_N1, making E1 a first-order reaction whose rate depends solely on [R – L]. A closer look at the deprotonation step is provided below.

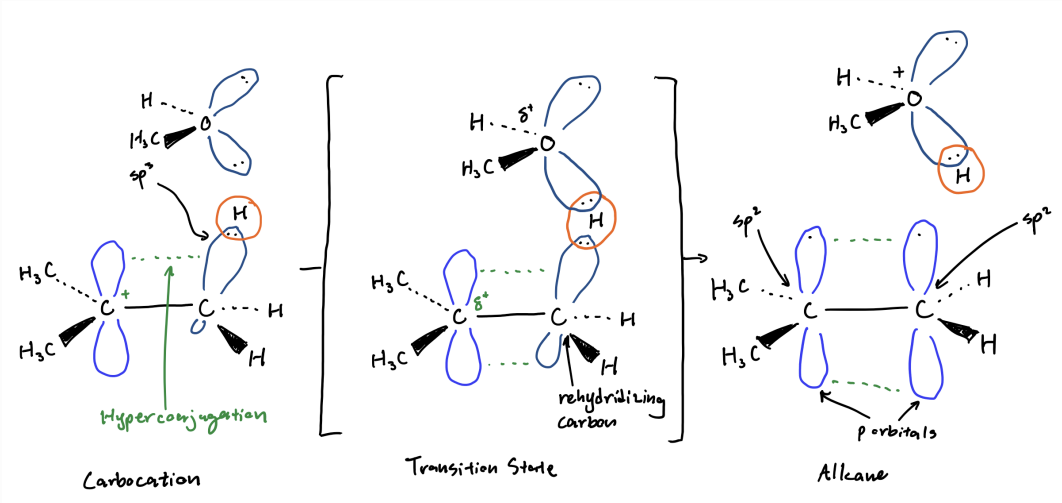

Figure 13: E1 deprotonation

Since deprotonation can occur at any neighbouring β C-H, E1 generally yields a mixture of alkenes across all possible pathways. Rotation around the resulting π bond gives rise to different stereoisomers.

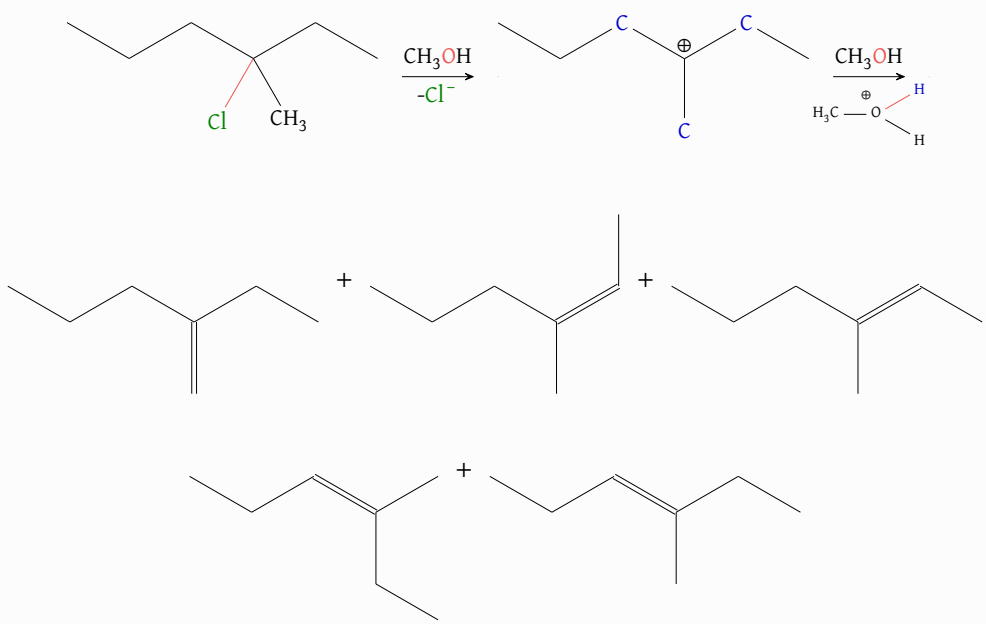

Figure 14: filler caption

Crucially, S_N1 and E1 share the exact same rate-determining step — spontaneous departure of the leaving group to form a carbocation. Since the leaving group has already departed by the time the carbocation partitions between substitution and elimination, the S_N1 **to E1 product ratio is independent of L**.

The ratio between E1 and S_N1 products is difficult to predict precisely, but E1 is generally favoured under two conditions.

(a) **Higher temperature**, as elimination carries a positive entropy change: HRL + B $\longrightarrow$ R + L + BH.

(b) A **poor nucleophile**, as S_N1 requires the reagent to form a new C-Nu bond, demanding greater nucleophilic strength. E1 is less sensitive to this since the reagent acts purely as a base, abstracting a surface H^+ rather than integrating into the molecular framework.

But what about **strong bases**?

Core Concept 2.3: Elimination 2 (E2)

Strong bases attack R – L **directly** at the β C-H, constituting the E2 mechanism. E2 is faster than S_N1 /E1 and faster than S_N2 when sterically hindered by α or β branching, as it condenses the distinct steps of E1 into a single concerted process.

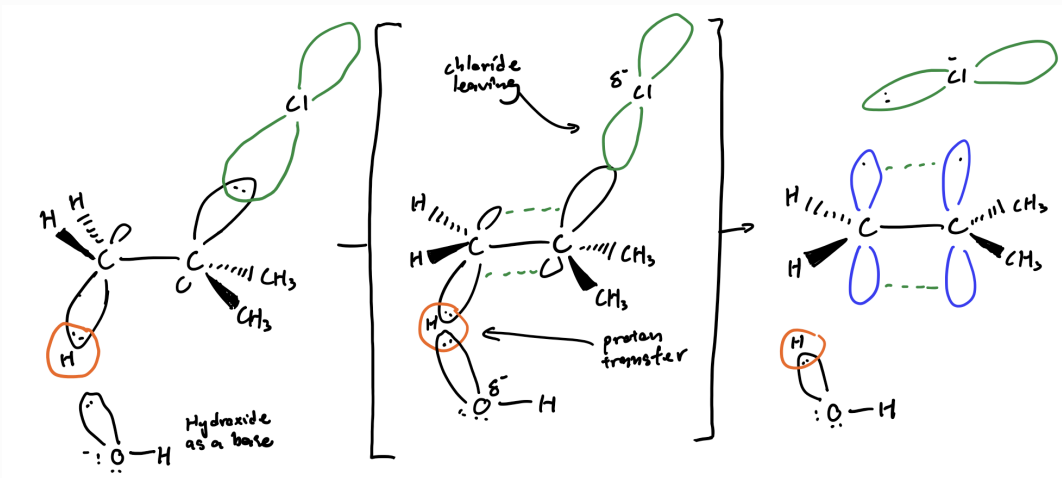

Figure 15: E2 transition state

Unlike E1, E2 is **second order** because the transition state involves both the substrate and the base.

$$Rate = k[\text{R–L}][\text{B}^-]$$

Consequently, a better leaving group and a stronger base both accelerate the reaction.

$$\text{RCl} < \text{RBr} < \text{RI}$$

Additionally, electron-withdrawing substituents within the substrate accelerate E2 by increasing the acidity of the target β C-H through withdrawal of its e^- density.

the EN from F makes H more acidic

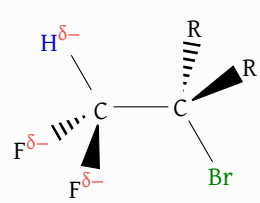

Figure 16: E2 hydrogen acidity amplification

E2 is also **stereospecific** — a given diastereomer of R – L yields only one stereoisomer of the alkene product. For instance, an RR/SS mixture yields exclusively one product, while an RS/SR mixture yields exclusively another.

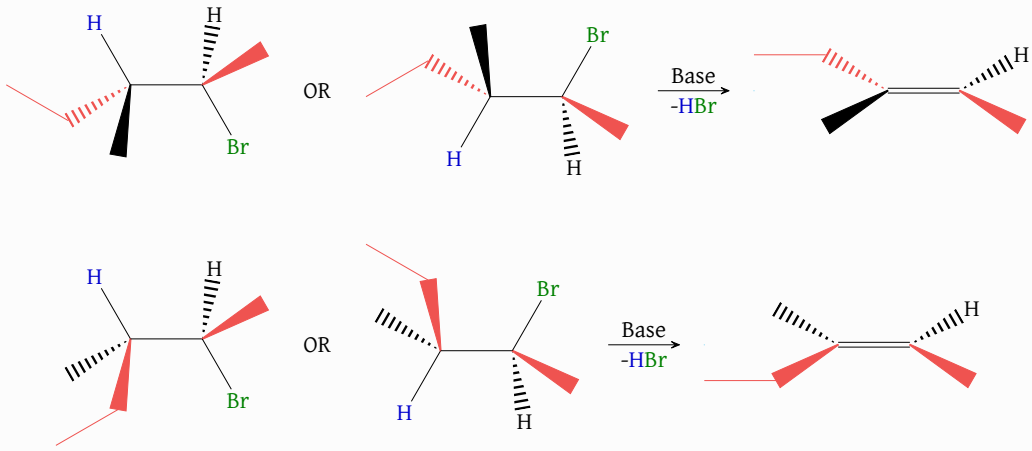

Figure 17: Stereospecific E2 outcomes; *RR (top left), SS (top right), SR (bottom left), RS (bottom right)*

This stereospecificity arises because H^+ and L depart simultaneously, requiring their initial positions to be coplanar to permit proper p orbital alignment in the forming π bond. This coplanarity can be achieved through either **anti**-periplanar (*trans*) or **syn**-periplanar (*cis*) geometry. When the relevant C-C bond can rotate freely, anti-periplanar geometry is strongly preferred as it minimises steric repulsion between the leaving group and the H – B transition complex.

In cases where free rotation is restricted — such as in cycloalkanes — syn-periplanar geometry must be adopted instead. This proceeds at a slower rate due to the increased steric hindrance between the leaving group and the H – B transition complex.

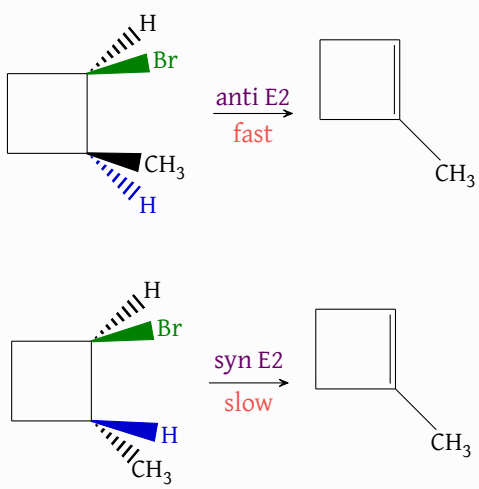

Figure 18: Anti and syn-periplaner E2 reaction rates

3 Competition

Having established the mechanics and characteristics of S_N2, S_N1, E1, and E2 reactions, it is important to recognise that all four pathways can be simultaneously possible across many overlapping conditions — and thus compete with one another in a given reaction environment. The table below organises these competing factors.

Type of haloalkane	Poor Nu (e.g. H_2O)	Weakly basic, good Nu	Strongly basic, unhindered Nu	Strongly basic, hindered Nu
Methyl	N/A	S_N2	S_N2	S_N2
Primary				
Unhindered	N/A	S_N2	S_N2	E2
Branched	N/A	S_N2	E2	E2
Secondary	Slow S_N1, E1	S_N2	E2	E2
Tertiary	S_N1, E1	S_N1	E2	E2

Core Concept 3.1: Factors of Competition between S_N2, S_N1, E1, and E2

Three core factors govern the competition between substitution and elimination.

(a) The base strength of the nucleophile

(b) Steric hindrance around the reacting carbon

(c) Steric hindrance around the nucleophile, particularly when it is a strong base

Primary substrates: Under poor nucleophile conditions, neither E1 nor S_N1 are viable due to insufficient hyperconjugation to sustain a carbocation, and neither S_N2 nor E2 proceed effectively as both depend on nucleophile strength. Moving from weakly basic to strongly hindered nucleophile conditions, the dominant factor becomes steric hindrance — whether along the haloalkane itself via β branching, or around the nucleophile through size or solvation in protic solvents. As hindrance increases, E2 is progressively favoured over S_N2.

Secondary substrates: Hyperconjugation is sufficient to support carbocation formation, making S_N1 and E1 viable under poor nucleophile conditions. E1 tends to dominate over S_N1 here, as abstracting a β H^+ is less demanding than integrating a nucleophile into the molecular framework. For stronger nucleophile conditions, the inherent steric hindrance of the secondary substrate generally drives the system toward E2, though S_N2 remains competitive when the nucleophile is weakly basic — insufficiently basic to abstract the β H^+ required by the E2 transition state.

Tertiary substrates: Steric hindrance completely shuts down S_N2, making S_N1 dominant under weakly basic, good nucleophile conditions. E2 dominates under strongly basic conditions, as it does for secondary substrates. Under poor nucleophile conditions, the S_N1 to E1 ratio becomes more balanced relative to secondary substrates — the greater stability of the tertiary carbocation lowers the activation energy for S_N1, allowing it to compete more effectively with E1.

Generally speaking, rather than memorizing each of these scenarios and their corresponding dominant mechanism, it's helpful to use S_N2 as a starting point and ask yourself the following questions.

(a) Is the nucleophile or haloalkane hindered? *If yes, the S_N2 reaction is likely not favored.*

(b) Is the nucleophile strongly basic? *If yes, E2. If no, S_N1/E1.*

However, keep in mind that there are other nuanced factors to consider such as absence of E1 preference for weakly basic, good nucleophile conditions.